ASSIGNMENT 8

HTML & CSS — Beginner-Friendly Solutions

All text is black. Each question has its own index.html and style.css file. The code uses simple HTML and CSS, without JavaScript or frameworks.

Question 1: Pseudo-Classes

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Pseudo-Classes</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="container">
    <h1>Pseudo-Classes Demo</h1>
    <div class="examples">
      <section><a href="#">Hover over this link</a><p><code>:hover</code> changes the link color.</
p></section>
      <section><button>Click or focus me</button><p><code>:focus</code> adds an outline; <code>:
active</code> changes it while pressed.</p></section>
      <section><ul><li>First item</li><li>Second item</li><li>Third item</li></ul><p><code>:nth-
child(2)</code> styles the second item.</p></section>
    </div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Basic page styling */
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f4f6f8; color: #222; }
.container { max-width: 1100px; margin: auto; padding: 32px 20px; text-align: center; }
.examples { display: flex; justify-content: center; align-items: stretch; gap: 20px; flex-wrap:
wrap; }
section { background: white; padding: 22px; border-radius: 10px; box-shadow: 0 3px 12px #0001; flex:
 1 1 250px; }
a { color: #6b3fa0; }
a:hover { color: #b04435; background: #fff0e9; }
button { padding: 10px 18px; border: 0; border-radius: 6px; background: #287a68; color: white;
cursor: pointer; }
button:focus { outline: 3px solid #e6a23c; outline-offset: 3px; }
button:active { background: #174b41; transform: scale(.96); }
li { padding: 8px; list-style-position: inside; }
li:nth-child(2) { background: #e6f2ff; color: #174b80; }
p { line-height: 1.5; }
```

Question 2: Tooltip Demonstration

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>CSS Tooltips</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>CSS Tooltip Demo</h1>
    <div class="items">
      <span class="tip">Home<span class="tooltip">Go to the home page</span></span>
      <span class="tip">Help<span class="tooltip">Find help and support</span></span>
      <span class="tip">Profile<span class="tooltip">View your profile</span></span>
    </div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Center tooltip examples with Flexbox */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; font-family: Arial, sans-serif; background: #f5f2ec; }
.page { min-height: 100vh; display: flex; flex-direction: column; justify-content: center; align-
items: center; padding: 24px; }
.items { display: flex; justify-content: center; gap: 28px; flex-wrap: wrap; }
.tip { position: relative; padding: 16px 22px; background: #315c66; color: white; border-radius:
8px; cursor: pointer; }
.tooltip { position: absolute; bottom: calc(100% + 12px); left: 50%; transform: translateX(-50%);
width: 190px; padding: 10px; border-radius: 6px; background: #222; color: white; text-align: center;
 font-size: 14px; opacity: 0; visibility: hidden; transition: opacity 0.3s ease-in-out; }
.tooltip::after { content: ""; position: absolute; top: 100%; left: 50%; border: 7px solid
transparent; border-top-color: #222; transform: translateX(-50%); }
.tip:hover .tooltip, .tip:focus-within .tooltip { opacity: 1; visibility: visible; }
```

Question 3: Image Showcase

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>Image Showcase</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>Image Showcase</h1>
    <div class="gallery">
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1500530855697-
b586d89ba3ee?w=700" alt="Mountain landscape"></div><figcaption>Mountain View</figcaption></figure>
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1470770841072-
f978cf4d019e?w=700" alt="Lake and mountains"></div><figcaption>Quiet Lake</figcaption></figure>
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1501785888041-
af3ef285b470?w=700" alt="Green hills"></div><figcaption>Green Hills</figcaption></figure>
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1441974231531-
c6227db76b6e?w=700" alt="Forest"></div><figcaption>Forest Trail</figcaption></figure>
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1470252649378-
9c29740c9fa8?w=700" alt="Sunrise"></div><figcaption>Golden Sunrise</figcaption></figure>
      <figure><div class="photo"><img src="https://images.unsplash.com/photo-1507525428034-
b723cf961d3e?w=700" alt="Ocean beach"></div><figcaption>Ocean Beach</figcaption></figure>
    </div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Responsive image grid */
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f3f5f7; color: #222; }
.page { max-width: 1200px; margin: auto; padding: 28px 20px; text-align: center; }
.gallery { display: grid; grid-template-columns: repeat(3, 1fr); gap: 22px; }
figure { margin: 0; background: white; padding: 10px; border-radius: 12px; box-shadow: 0 4px 14px
#0002; overflow: hidden; }
.photo { position: relative; overflow: hidden; border-radius: 9px; }
.photo img { display: block; width: 100%; height: 190px; object-fit: cover; transition: transform .
3s ease; }
.photo::after { content: ""; position: absolute; inset: 0; background: #0005; opacity: 0;
transition: opacity .3s ease; }
figure:hover img { transform: scale(1.1); }
figure:hover .photo::after { opacity: 1; }
figcaption { padding: 12px 4px 4px; font-weight: bold; }
@media (max-width: 700px) { .gallery { grid-template-columns: 1fr; } }
```

Question 4: CSS Selectors Practice

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>CSS Selectors</title><link rel="stylesheet" href="style.css"></head>
<body>
  <header><h1>CSS Selectors Practice</h1></header>
  <main>
    <section class="intro">
      <p>This paragraph demonstrates an element selector.</p>
      <p class="notice">This paragraph uses a class selector.</p>
      <p id="special">This paragraph uses an ID selector.</p>
      <a href="https://example.com" target="_blank">Attribute selector link</a>
    </section>
    <section class="examples">
      <h2>Combinator examples</h2>
      <div class="group">
        <p>This is a descendant paragraph.</p>
        <ul><li>Child list item one</li><li>Child list item two</li></ul>
      </div>
      <h3>Adjacent heading</h3>
      <p>This paragraph comes immediately after h3.</p>
      <p>This is a later sibling paragraph.</p>
    </section>
  </main>
  <footer><p>CSS selectors help target specific HTML elements.</p></footer>
</body>
</html>
```

CSS file — save as style.css

```
/* Element selector: styles all paragraphs */
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; color: #242424; line-height: 1.6; }
header, footer { background: #394b59; color: white; text-align: center; padding: 18px; }
main { max-width: 850px; margin: auto; padding: 24px; }
p { margin: 12px 0; }
/* Class selector */
.notice { background: #fff1c9; padding: 10px; }
/* ID selector */
#special { color: #8c3c30; font-weight: bold; }
/* Attribute selector */
a[target="_blank"] { color: #176b87; text-decoration: underline; }
/* Descendant combinator: any p inside .group */
.group p { background: #eef4f7; padding: 10px; }
/* Child combinator: direct li children of ul */
ul > li { margin: 6px 0; }
/* Adjacent sibling: p directly after h3 */
h3 + p { color: #287a68; font-weight: bold; }
/* General sibling: all p siblings after h3 */
h3 ~ p { border-left: 3px solid #d9a441; padding-left: 10px; }
```

Question 5: Styled Button Collection

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>Button Collection</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>Styled Button Collection</h1>
    <div class="buttons">
      <button class="primary">Primary</button>
      <button class="secondary">Secondary</button>
      <button class="outline">Outline</button>
      <button disabled>Disabled</button>
    </div>
    <p>Use Tab to see the focus outline. Disabled buttons cannot be clicked.</p>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Consistent button sizing and spacing */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; font-family: Arial, sans-serif; background: #f6f4f0; }
.page { min-height: 100vh; display: flex; flex-direction: column; align-items: center; justify-
content: center; padding: 24px; text-align: center; }
.buttons { display: flex; justify-content: center; gap: 16px; flex-wrap: wrap; }
button { padding: 12px 24px; border-radius: 8px; font-size: 16px; cursor: pointer; transition: .25s
ease-in-out; box-shadow: 0 3px 8px #0002; }
.primary { background: #316b5c; color: white; border: 2px solid #316b5c; }
.secondary { background: #e6b85c; color: #222; border: 2px solid #e6b85c; }
.outline { background: transparent; color: #315c66; border: 2px solid #315c66; }
button:hover:not(:disabled) { transform: translateY(-3px); box-shadow: 0 6px 12px #0003; }
button:active:not(:disabled) { transform: scale(.96); }
button:focus { outline: 3px solid #b65c3a; outline-offset: 3px; }
button:disabled { background: #d5d5d5; color: #777; border: 2px solid #c4c4c4; cursor: not-allowed;
box-shadow: none; }
```

Question 6: CSS Animation Demonstration

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>CSS Animations</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>CSS Animation Demo</h1>
    <div class="stage">
      <div class="moving">Moving Box</div>
      <div class="ball"></div>
      <p class="color-text">Color Changing Text</p>
    </div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Keep animation examples centered */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; font-family: Arial, sans-serif; background: #f5f2ec; }
.page { min-height: 100vh; display: flex; flex-direction: column; align-items: center; justify-
content: center; padding: 24px; }
.stage { width: min(650px, 100%); min-height: 300px; display: flex; flex-direction: column; align-
items: center; justify-content: space-evenly; gap: 24px; background: white; border-radius: 14px;
box-shadow: 0 4px 15px #0002; padding: 24px; }
.moving { padding: 14px 22px; color: white; background: #376b80; border-radius: 8px; animation:
move 3s ease-in-out infinite alternate; }
.ball { width: 42px; height: 42px; border-radius: 50%; background: #d98245; animation: bounce 1.2s
ease-in-out infinite; }
.color-text { font-size: 20px; font-weight: bold; animation: colorChange 3s ease-in-out infinite;
animation-delay: .4s; }
@keyframes move { from { transform: translateX(-120px); } to { transform: translateX(120px); } }
@keyframes bounce { 0%, 100% { transform: translateY(0); } 50% { transform: translateY(-35px); } }
@keyframes colorChange { 0%, 100% { color: #376b80; } 50% { color: #b34c3c; } }
```

Question 7: User Interface Components

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>UI Components</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="card">
    <h1>Quick Search</h1>
    <label for="search">Search items</label>
    <input id="search" type="search" placeholder="Type to search...">
    <label for="category">Choose a category</label>
    <select id="category"><option>All categories</option><option>Books</option><option>Courses</
option><option>Tools</option></select>
    <div class="actions"><button>Search</button><button class="clear">Clear</button></div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Soft gradient background and card layout */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; display: flex; justify-content: center; align-items: center;
padding: 20px; font-family: Arial, sans-serif; background: linear-gradient(135deg, #f4e7d3,
#dce9e7); }
.card { width: min(430px, 100%); display: flex; flex-direction: column; gap: 12px; padding: 28px;
background: white; border-radius: 14px; box-shadow: 0 8px 24px #0002; }
input, select { width: 100%; padding: 12px; border: 1px solid #aaa; border-radius: 7px; font-size:
16px; background: white; }
input:focus, select:focus { outline: 3px solid #9ac4bd; border-color: #316b5c; }
.actions { display: flex; gap: 12px; margin-top: 8px; }
button { flex: 1; padding: 12px; border: 0; border-radius: 7px; background: #316b5c; color: white;
cursor: pointer; transition: .2s; }
button:hover { background: #244f45; }
button:active { transform: scale(.97); }
button.clear { background: #e6b85c; color: #222; }
button.clear:hover { background: #d5a343; }
```

Question 8: Box-Sizing Comparison

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>Box Sizing Comparison</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>Box-Sizing Comparison</h1>
    <div class="comparison">
      <section><h2>Content-box</h2><div class="box content-box">Width: 250px<br>Padding: 25px</
div><p>The default adds padding and border outside the set width. Total width is 250 + 50 + borders.
</p></section>
      <section><h2>Border-box</h2><div class="box border-box">Width: 250px<br>Padding: 25px</
div><p><code>border-box</code> includes padding and border inside the set width. Total width stays
250px.</p></section>
    </div>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Compare equal declared widths and padding */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; font-family: Arial, sans-serif; background: #f4f6f8; color:
#222; }
.page { max-width: 950px; margin: auto; padding: 28px 20px; text-align: center; }
.comparison { display: flex; justify-content: center; align-items: stretch; gap: 24px; flex-wrap:
wrap; }
section { flex: 1 1 280px; padding: 20px; background: white; border-radius: 12px; box-shadow: 0 4px
14px #0002; }
.box { width: 250px; max-width: 100%; padding: 25px; margin: 18px auto; border: 5px solid #222;
color: white; font-weight: bold; }
.content-box { box-sizing: content-box; background: #a85b45; }
.border-box { box-sizing: border-box; background: #357b70; }
p { line-height: 1.5; }
@media (max-width: 600px) { .comparison { flex-direction: column; } }
```

Question 9: Combined CSS Effects

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>Combined CSS Effects</title><link rel="stylesheet" href="style.css"></head>
<body>
  <main class="page">
    <h1>Combined CSS Effects</h1>
    <div class="examples">
      <button class="effect one">Explore</button>
      <button class="effect two">Learn More</button>
      <a class="effect link" href="#">Visit Link</a>
    </div>
    <p>Hover, focus, or press each element to see its effect.</p>
  </main>
</body>
</html>
```

CSS file — save as style.css

```
/* Pseudo-classes, transitions, and transforms together */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; font-family: Arial, sans-serif; background: #f5f2ec; }
.page { min-height: 100vh; display: flex; flex-direction: column; align-items: center; justify-
content: center; padding: 24px; text-align: center; }
.examples { display: flex; justify-content: center; align-items: center; gap: 22px; flex-wrap: wrap;
 }
.effect { display: inline-block; padding: 14px 24px; border: 0; border-radius: 9px; font: inherit;
text-decoration: none; cursor: pointer; box-shadow: 0 3px 8px #0002; transition: transform .25s
ease-in-out, background .25s ease-in-out, box-shadow .25s ease-in-out; }
.one { background: #356b80; color: white; }
.two { background: #dba94d; color: #222; }
.link { background: #54775b; color: white; }
.effect:hover { transform: scale(1.08); background: #9c5945; color: white; box-shadow: 0 8px 18px
#0003; }
.effect:focus { outline: 3px solid #222; outline-offset: 4px; }
.effect:active { transform: scale(.96); }
```

Question 10: Mini User Interface Layout

HTML file — save as index.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.
0"><title>Mini UI Layout</title><link rel="stylesheet" href="style.css"></head>
<body>
  <header><h1>Study Space</h1></header>
  <main>
    <section class="card">
      <img src="https://images.unsplash.com/photo-1455390582262-
044c?auto=format&fit=crop&w=700&q=80" alt="Notebook and study desk">
      <div class="content">
        <h2>Join Our Study List</h2>
        <p>Get simple study tips and updates.</p>
        <form>
          <label for="name">Your name</label>
          <input id="name" type="text" placeholder="Enter your name" required>
          <label for="email">Email address</label>
          <input id="email" type="email" placeholder="Enter your email" required>
          <div class="buttons"><button type="submit">Join Now</button><button type="reset"
class="reset">Reset</button></div>
        </form>
      </div>
    </section>
  </main>
  <footer><p>Made for learning HTML and CSS</p></footer>
</body>
</html>
```

CSS file — save as style.css

```
/* Shared sizing and responsive mini interface */
* { box-sizing: border-box; }
body { margin: 0; min-height: 100vh; display: flex; flex-direction: column; font-family: Arial,
sans-serif; background: #f4f1eb; color: #222; }
header, footer { text-align: center; padding: 16px; background: #365c59; color: white; }
main { flex: 1; display: flex; justify-content: center; align-items: center; padding: 24px; }
.card { width: min(760px, 100%); display: flex; gap: 24px; align-items: center; padding: 22px;
background: white; border: 1px solid #ddd; border-radius: 14px; box-shadow: 0 5px 18px #0002; }
.card img { width: 42%; max-height: 320px; object-fit: cover; border-radius: 10px; }
.content { flex: 1; }
form { display: flex; flex-direction: column; gap: 9px; }
input { padding: 11px; border: 1px solid #aaa; border-radius: 6px; font-size: 15px; }
input:focus { outline: 3px solid #a9c9bf; border-color: #365c59; }
.buttons { display: flex; gap: 10px; margin-top: 8px; }
button { flex: 1; padding: 11px; border: 0; border-radius: 6px; background: #365c59; color: white;
cursor: pointer; transition: .2s; }
button:hover { background: #24413f; transform: translateY(-2px); }
button:focus { outline: 3px solid #dba94d; outline-offset: 2px; }
button:active { transform: scale(.97); }
button.reset { background: #dba94d; color: #222; }
button.reset:hover { background: #c59435; }
@media (max-width: 650px) { .card { flex-direction: column; } .card img { width: 100%; max-height:
220px; } }
```